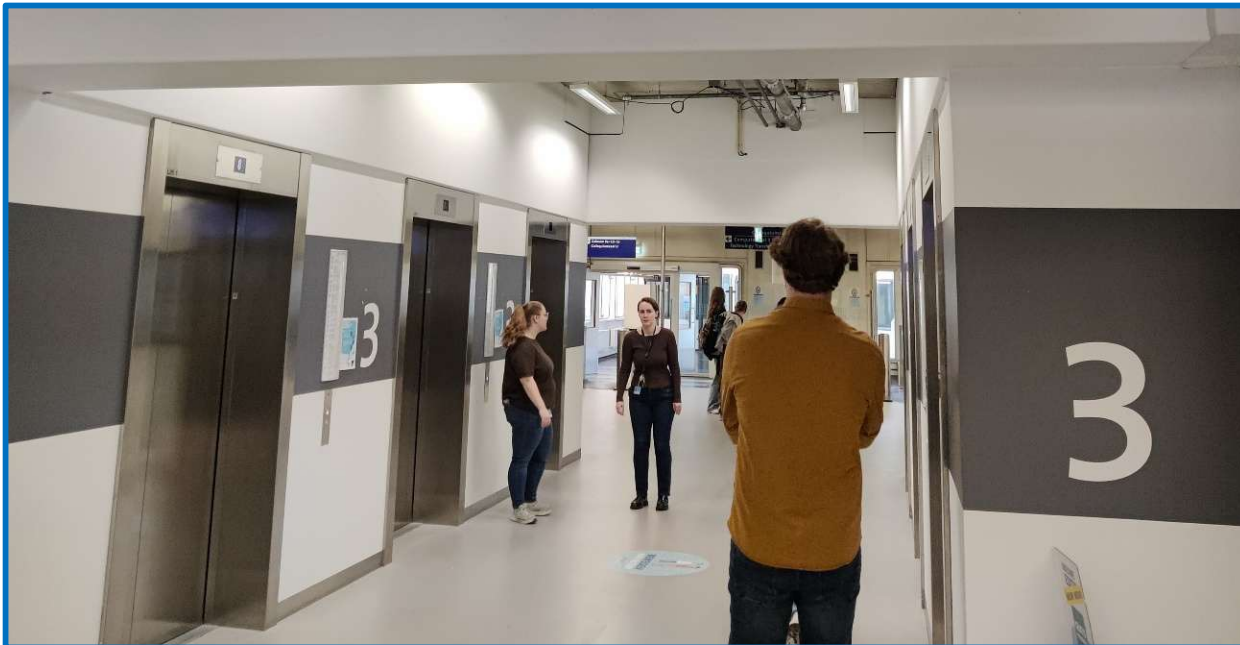
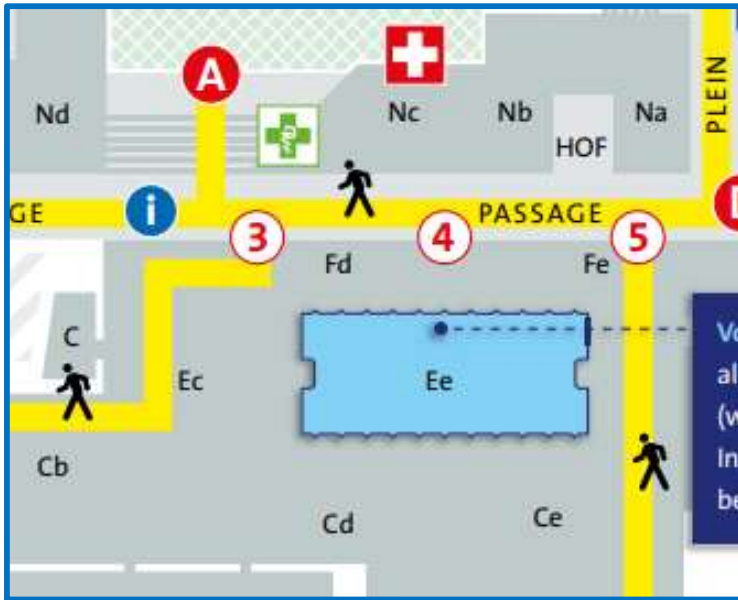


Wednesday September 30
13:30 – 16:30
Room COO-3



Name:

Student number:



**Master of Science
in
Molecular Medicine**



MM-MBC-A

Molecular Biology of the Cell – part A

Exam

Tuesday, October 6, 2020

13:00 – 16:00 hrs.

Onderwijsruimte 39

FIRST READ THE INSTRUCTIONS ON PAGE 2

**Wednesday September 30
13:30 – 16:30
Room COO-3**

Name:

Student number:

INSTRUCTIONS

- This exam contains 8 questions.
For each question, a maximum of 10 points can be scored, resulting in a maximum score of 80 points.
 - 44 points will result in a score of 5.5
 - 80 points will result in a score of 10
- Please be brief, but not too brief:
answering a question with a simple 'yes' or 'no' is not sufficient, unless specifically indicated
- Always explain your answer, unless specifically indicated
- Make sure your handwriting is legible
- Do not write answers on the reverse side of the pages
- Write your name on all pages

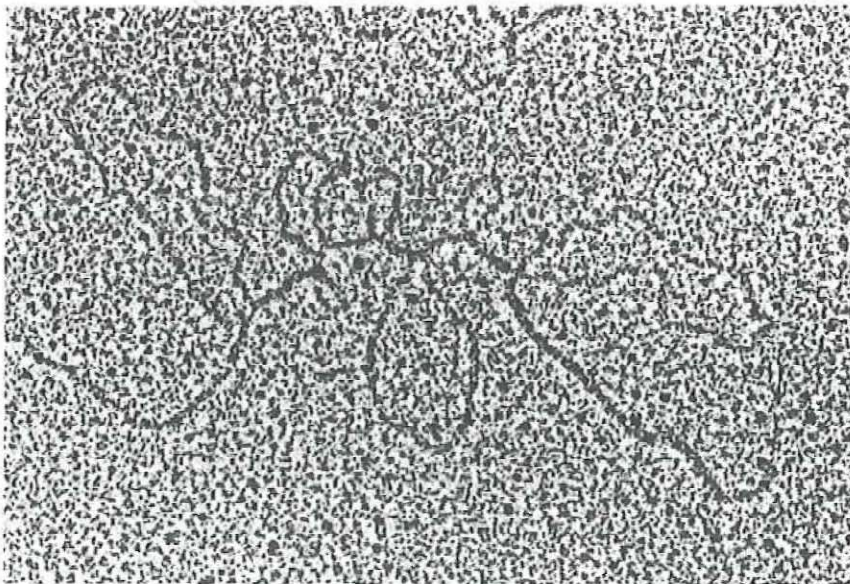
The intron-exon structure of eukaryotic genes came as a shock.

In the early, skeptical days, the most convincing demonstration was visual, as shown, for example, by the electron micrograph below (left panel).

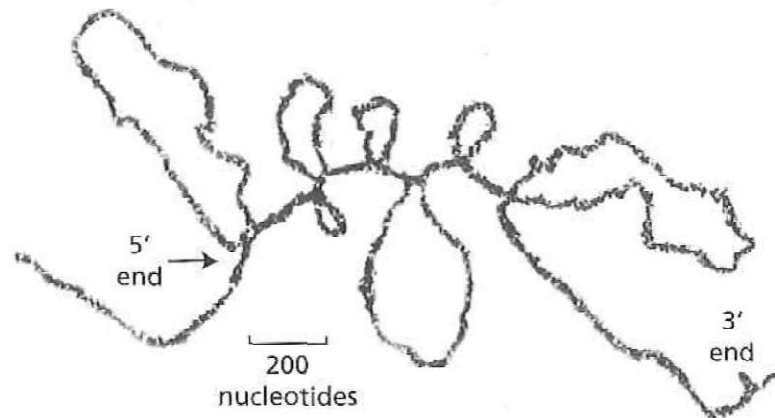
This image was obtained by hybridizing ovalbumin mRNA to a long segment of DNA that contains the gene. To those used to looking at single- and double-stranded nucleic acids in the electron microscope, the structure was clear: a set of single-stranded tails and loops emanating from a central duplex segment whose ends corresponded to the ends of the mRNA (below, right panel).

To the extent possible, describe the intron-exon structure of this gene.

(A) ELECTRON MICROGRAPH



(B) INTERPRETATION



Under stress conditions the levels of the p53 protein (tumor suppressor / DNA-binding protein) regulate cellular decisions about proliferation, apoptosis and senescence.

a) Arginine residues in the interface between p53 and DNA are frequently mutated in tumors. What type of interaction do you expect between an Arg and DNA?

Under stress conditions the levels of the p53 protein (tumor suppressor / DNA-binding protein) regulate cellular decisions about proliferation, apoptosis and senescence.

a) Arginine residues in the interface between p53 and DNA are frequently mutated in tumors. What type of interaction do you expect between an Arg and DNA?

b) What other amino acids could make this interaction?

Under stress conditions the levels of the p53 protein (tumor suppressor / DNA-binding protein) regulate cellular decisions about proliferation, apoptosis and senescence.

- a)** Arginine residues in the interface between p53 and DNA are frequently mutated in tumors. What type of interaction do you expect between an Arg and DNA?
- b)** What other amino acids could make this interaction?
- c)** A mutation could also affect the ability of p53 to fold into a three-dimensional structure and thereby affect its stability. Different classes of amino acids can cause different folding problems. Please discuss two different ways in which a mutation could cause a problem.

Under stress conditions the levels of the p53 protein (tumor suppressor / DNA-binding protein) regulate cellular decisions about proliferation, apoptosis and senescence.

- a)** Arginine residues in the interface between p53 and DNA are frequently mutated in tumors. What type of interaction do you expect between an Arg and DNA?
- b)** What other amino acids could make this interaction?
- c)** A mutation could also affect the ability of p53 to fold into a three-dimensional structure and thereby affect its stability. Different classes of amino acids can cause different folding problems. Please discuss two different ways in which a mutation could cause a problem.
- d)** A protease causes the degradation of other proteins to amino acids. What bonds will it break?

Under stress conditions the levels of the p53 protein (tumor suppressor / DNA-binding protein) regulate cellular decisions about proliferation, apoptosis and senescence.

- a)** Arginine residues in the interface between p53 and DNA are frequently mutated in tumors. What type of interaction do you expect between an Arg and DNA?
- b)** What other amino acids could make this interaction?
- c)** A mutation could also affect the ability of p53 to fold into a three-dimensional structure and thereby affect its stability. Different classes of amino acids can cause different folding problems. Please discuss two different ways in which a mutation could cause a problem.
- d)** A protease causes the degradation of other proteins to amino acids. What bonds will it break?
- e)** The proteasome has 14 subunits, at what level of folding do these come together?

Ribosomes are composed of one large and one small subunit.

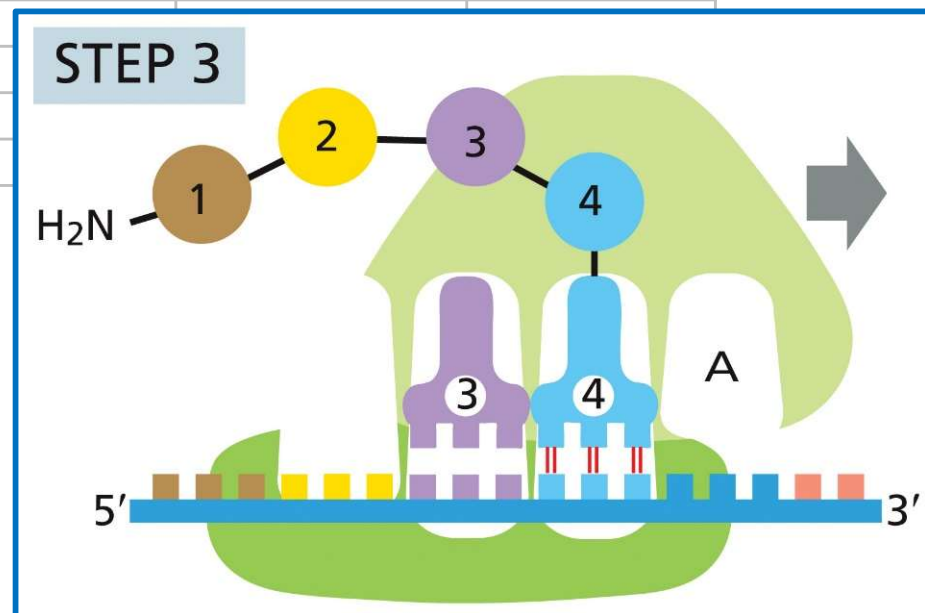
In the table below, indicate whether a characteristic is unique to the large subunit, or to the small subunit, or is common for both subunits.

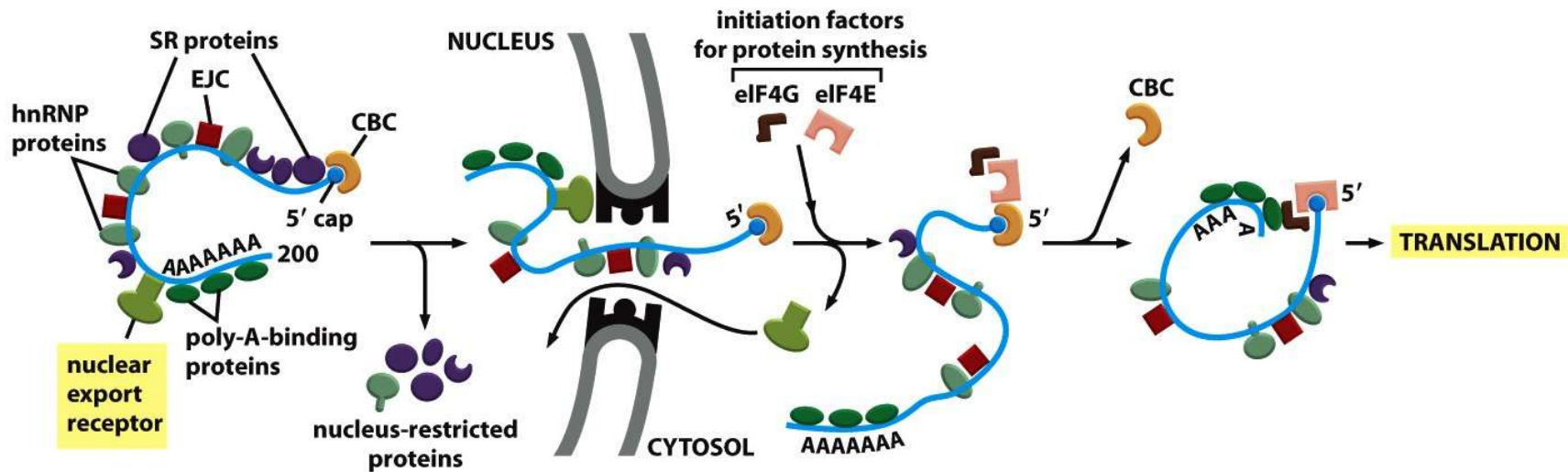
	Large subunit only	Small subunit only	Both subunits
Binds mRNA			
Build of rRNA			
Contains proteins			
Has catalytic activity			
Scans for start-codon			
Interacts with growing peptide			
Binding site for antibiotics			
Binds tRNA			
Slides first during elongation			
Is assembled in the nucle(ol)us			

Ribosomes are composed of one large and one small subunit.

In the table below, indicate whether a characteristic is unique to the large subunit, or to the small subunit, or is common for both subunits.

	Large subunit only	Small subunit only	Both subunits
Binds mRNA			
Build of rRNA			
Contains proteins			
Has catalytic activity			
Scans for start-codon			
Interacts with growing peptide			
Binding site for antibiotics			
Binds tRNA			
Slides first during elongation			
Is assembled in the nucle(ol)us			





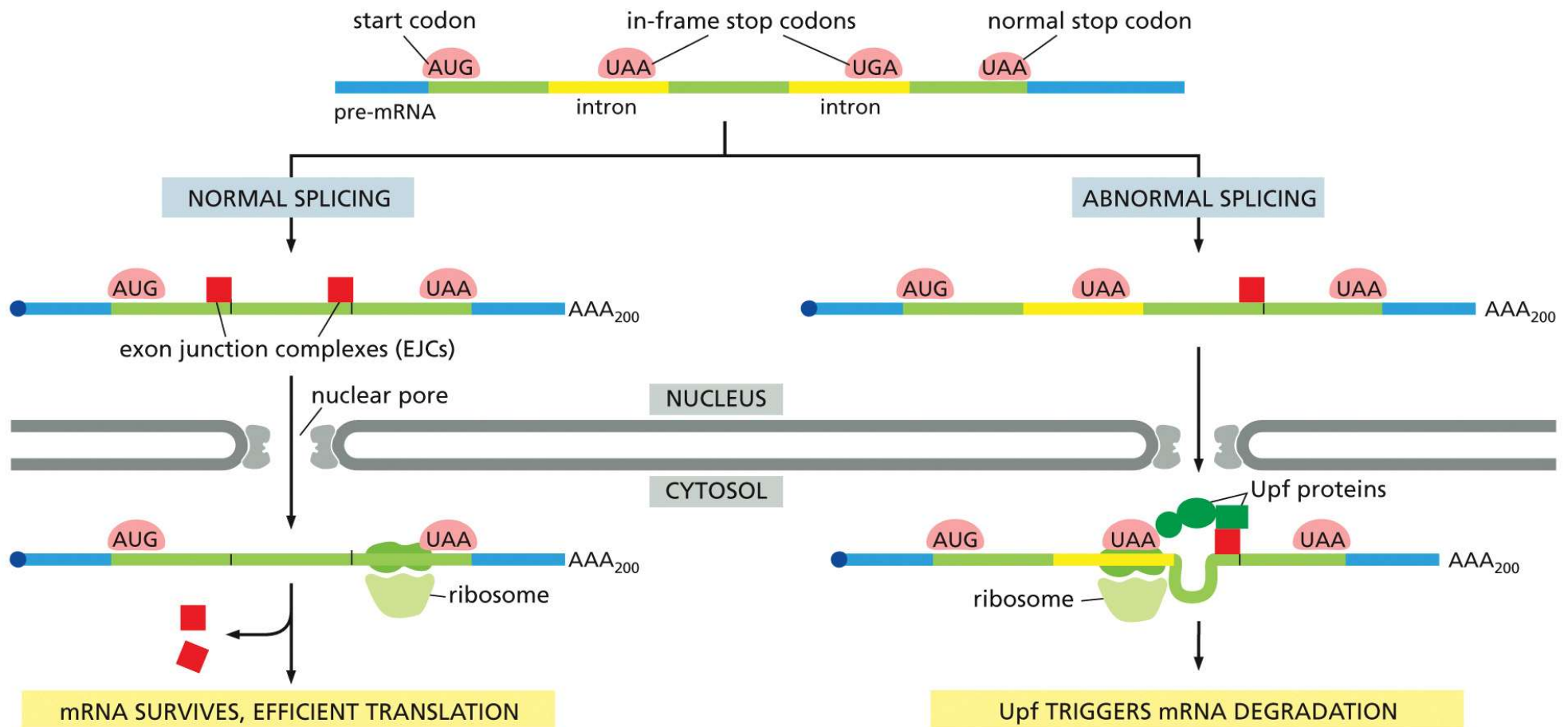
CBC: cap-binding complex

SR proteins: so-named because they contain a domain rich in serines and arginines, these proteins assemble on exon sequences

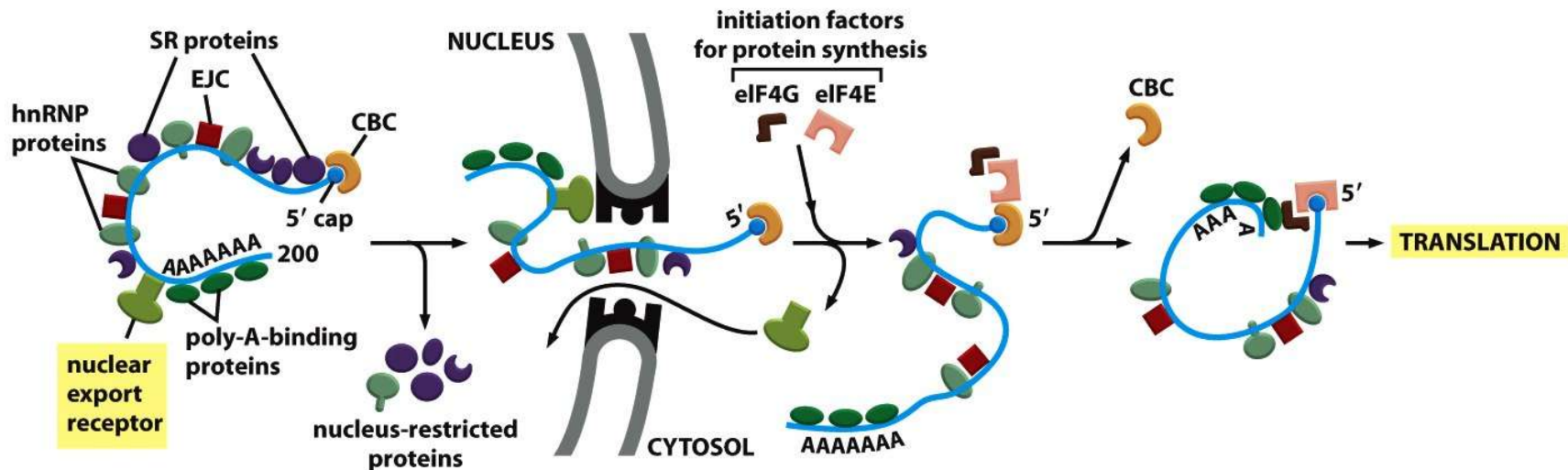
EJC: exon-junction complex

a) The EJC is not a nucleus-restricted protein complex. What is the function of the EJC in the cytoplasm?

Quality control: nonsense-mediated mRNA decay



Adapted from J. Lykke-Andersen et al., *Cell* 103:1121–1131, 2000.



CBC: cap-binding complex

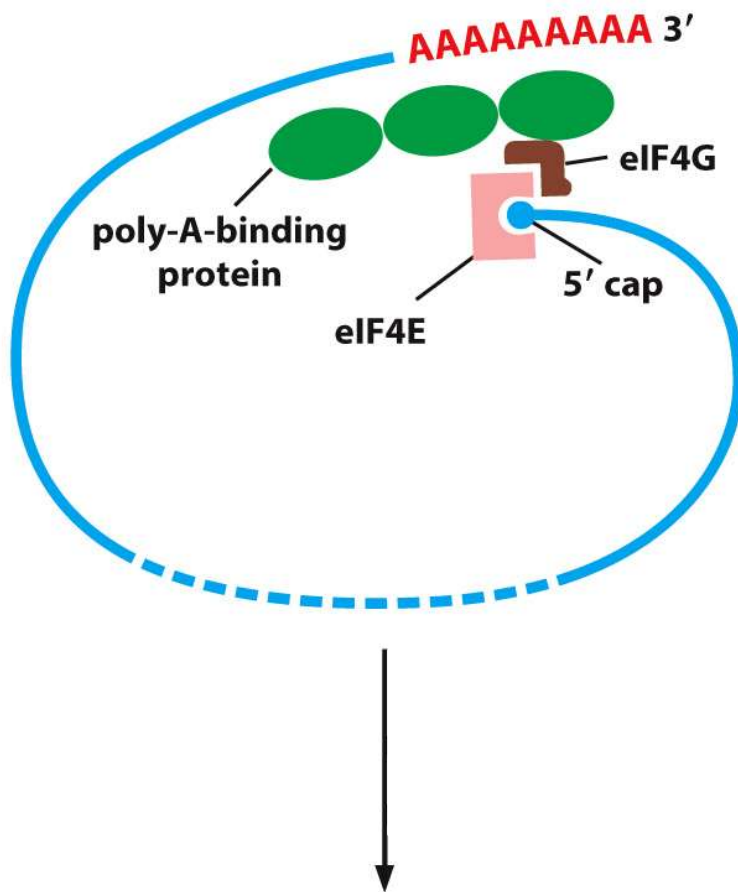
SR proteins: so-named because they contain a domain rich in serines and arginines, these proteins assemble on exon sequences

EJC: exon-junction complex

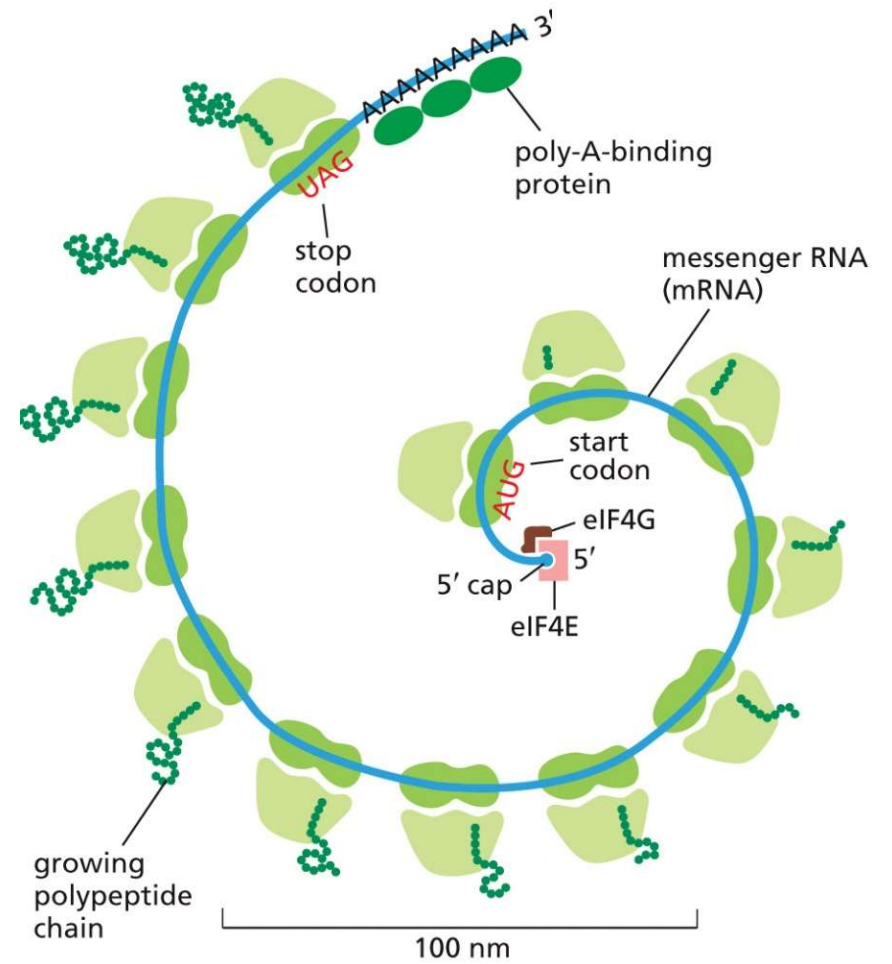
a) The EJC is not a nucleus-restricted protein complex. What is the function of the EJC in the cytoplasm?

b) Shortly before translation, the mRNA-protein complex is shown in the figure as a circular structure.

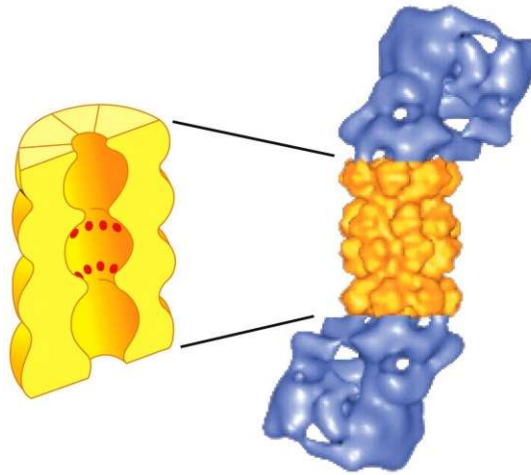
Does this have (a) biological function(s)? Explain.



TRANSLATION INITIATION



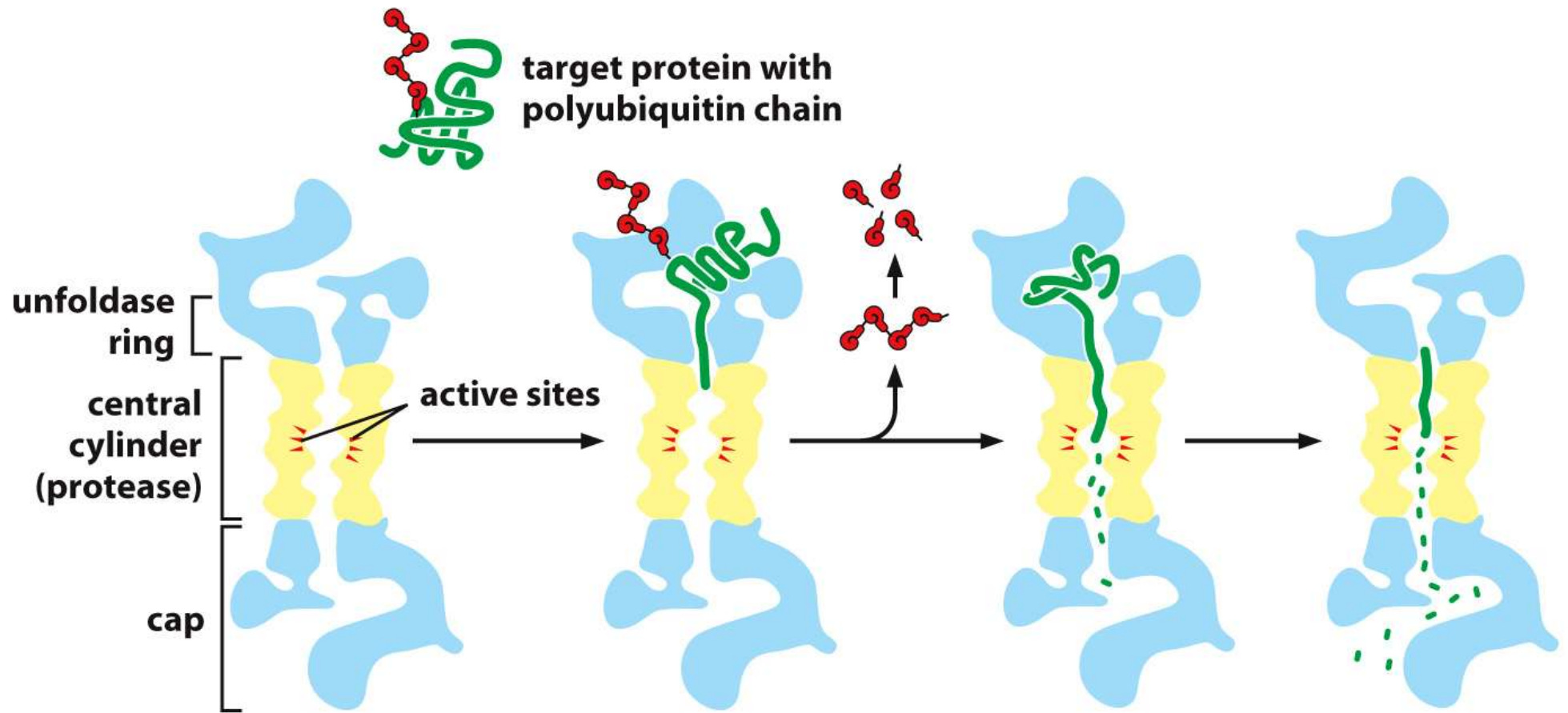
The figure shows the central cylinder of a proteasome (left) and the entire proteasome (right).



How are proteins targeted for degradation by proteasomes?

Describe the process by which a misfolded protein ends up in the proteasome.

The proteasome



Ubiquitin ligase

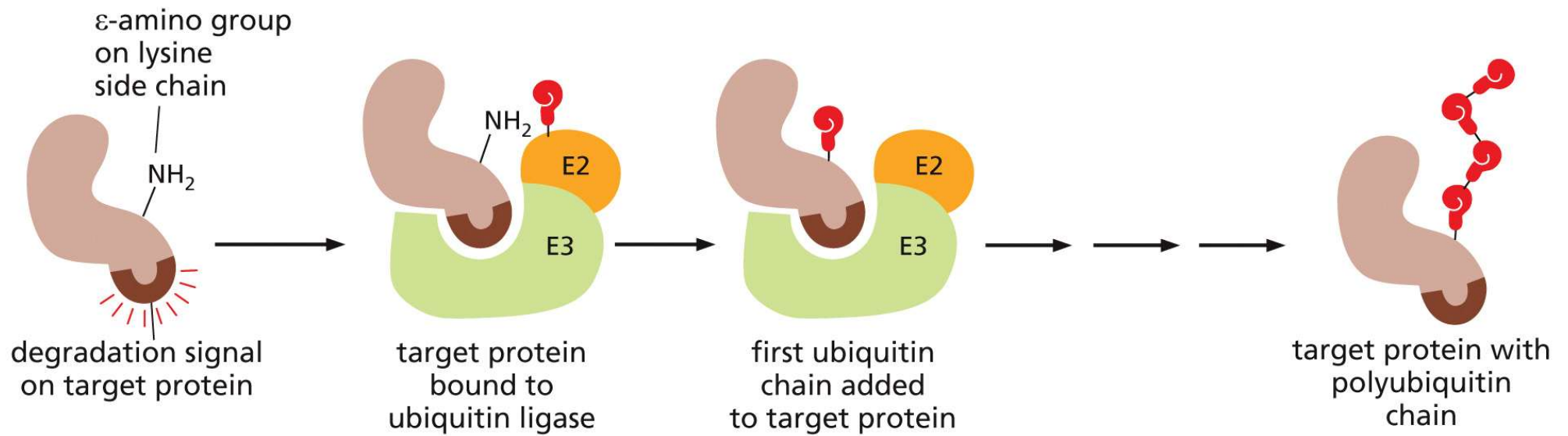
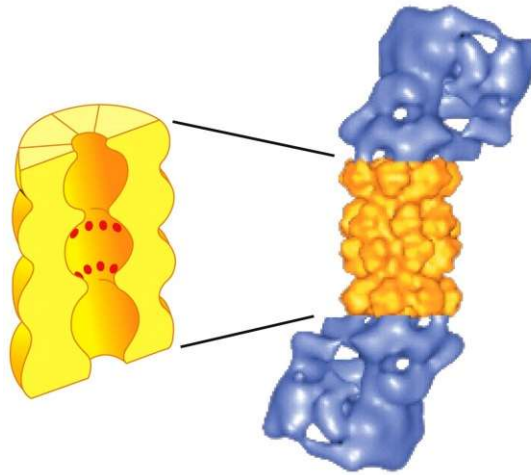


Figure 3-66B

The figure shows the central cylinder of a proteasome (left) and the entire proteasome (right).

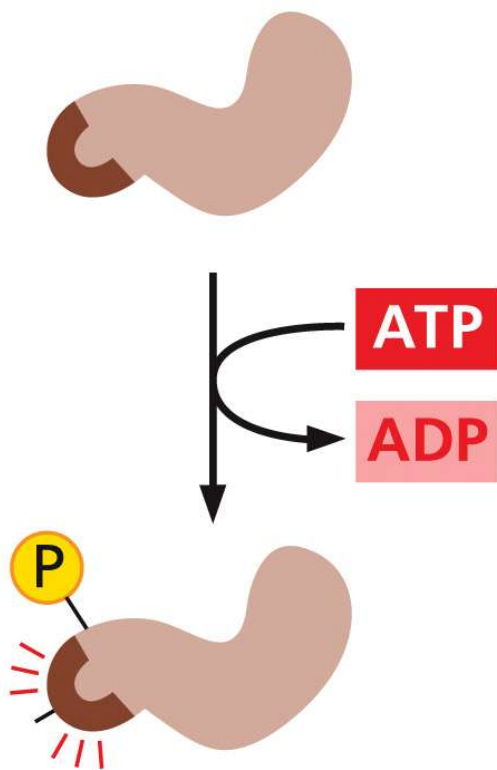


How are proteins targeted for degradation by proteasomes?

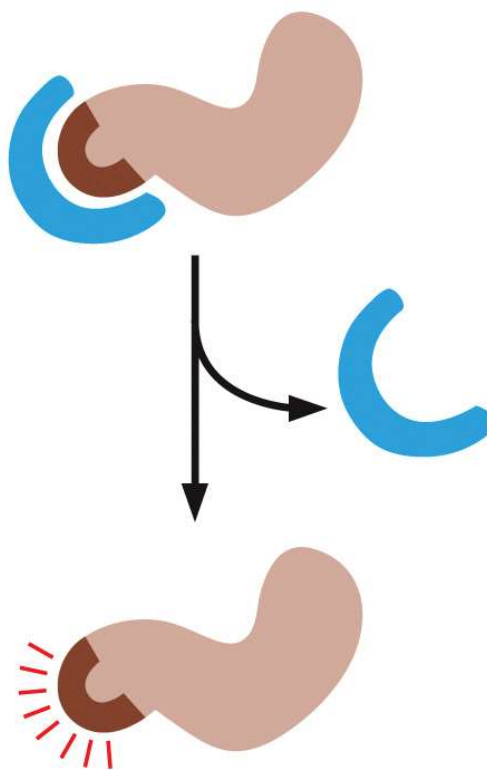
Describe the process by which a misfolded protein ends up in the proteasome.

Regulated destruction

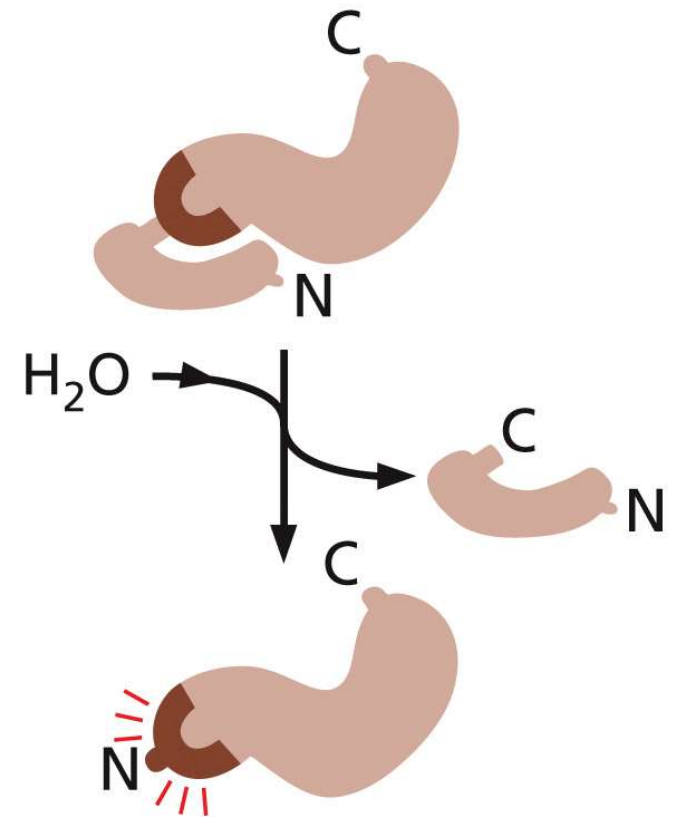
ACTIVATION OF A DEGRADATION SIGNAL



phosphorylation
by protein kinase

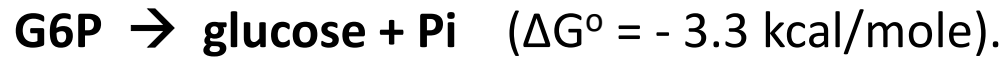


unmasking by
protein dissociation

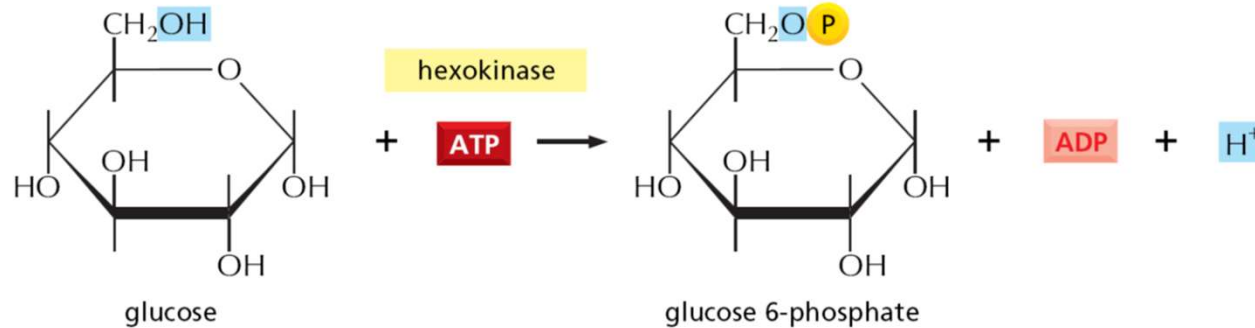
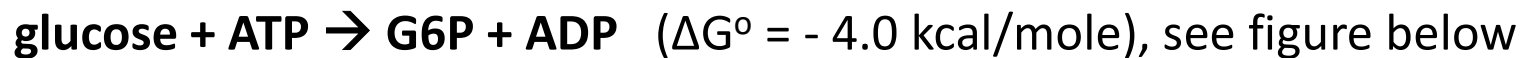


creation of destabilizing
N-terminus

The liver provides glucose to the rest of the body between meals. It does so by breaking down glycogen, forming glucose 6-phosphate which is subsequently converted to glucose by splitting of the phosphate by hydrolysis:



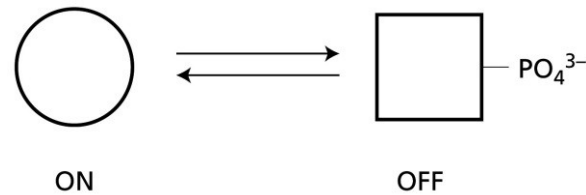
Conversion of glucose 6-phosphate to glucose could also occur by reversing the reaction



By the reversed reaction $\text{G6P} + \text{ADP} \rightarrow \text{glucose} + \text{ATP}$ the liver would generate both glucose and ATP!

Why does the liver remove the phosphate by hydrolysis rather than by the reaction which also generates ATP? Explain your answer.

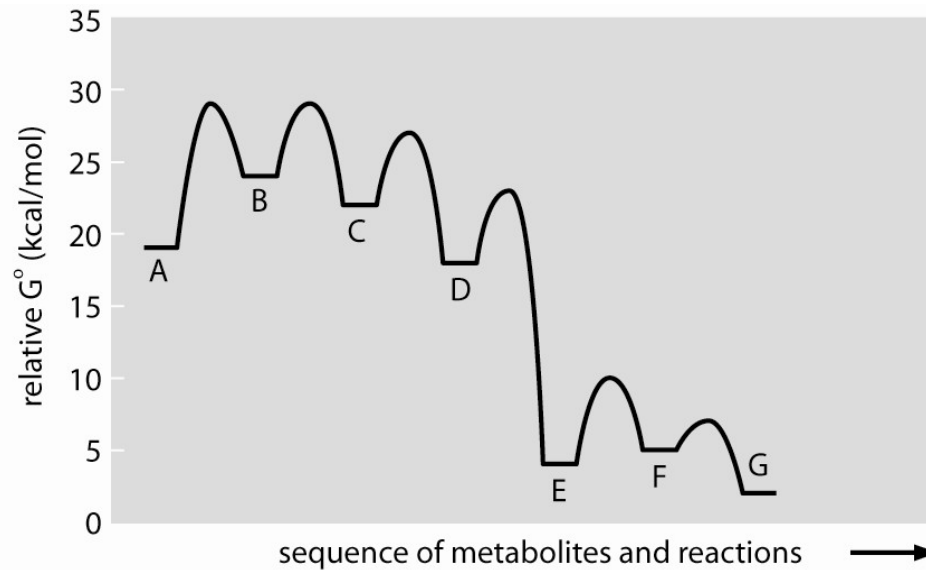
Many proteins inside cells are regulated by phosphorylation and dephosphorylation. Imagine a metabolic reaction that is catalyzed by an enzyme which is completely active when not phosphorylated and completely inactive when phosphorylated (see figure).



Inside the cell the rate of this metabolic reaction varies continuously between very fast and very slow (or zero) at a given substrate concentration.

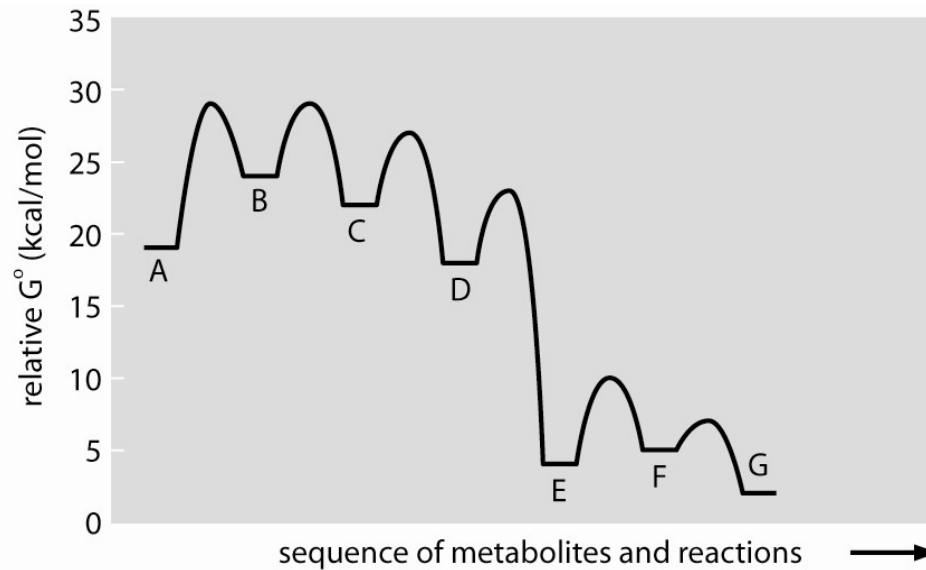
How can the reaction in the cell proceed at any rate between very fast and zero, when the enzyme can only be either on or off?

The energetic trajectory of a series of reactions in a catabolic pathway can be plotted on a graph like that in the figure. The graph shows a hypothetical pathway similar to glycolysis, in which the catabolic intermediates are designated A, B, C, etc. For the following questions, and throughout this exam, explain your answers.



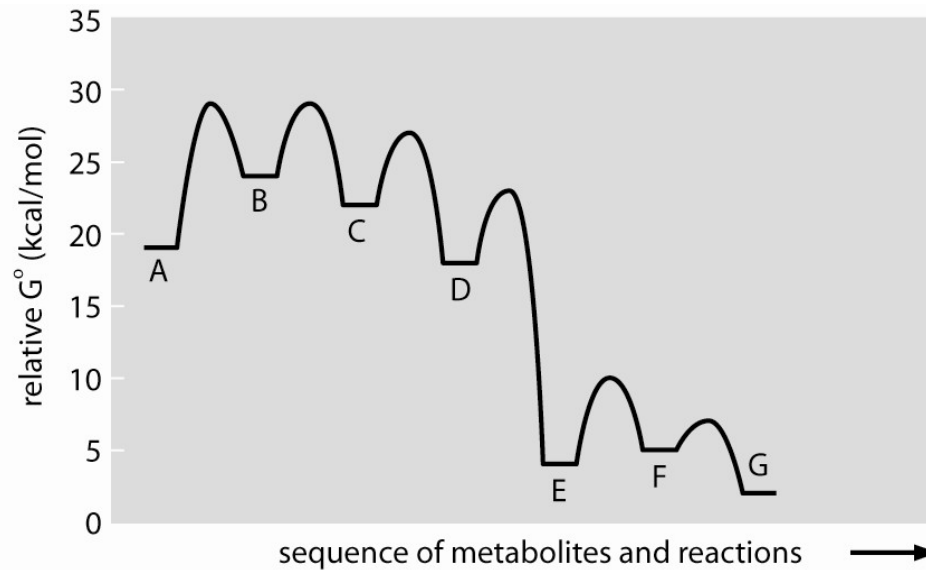
a) Which reaction(s) is most likely to occur without an enzyme in the cell?

The energetic trajectory of a series of reactions in a catabolic pathway can be plotted on a graph like that in the figure. The graph shows a hypothetical pathway similar to glycolysis, in which the catabolic intermediates are designated A, B, C, etc. For the following questions, and throughout this exam, explain your answers.



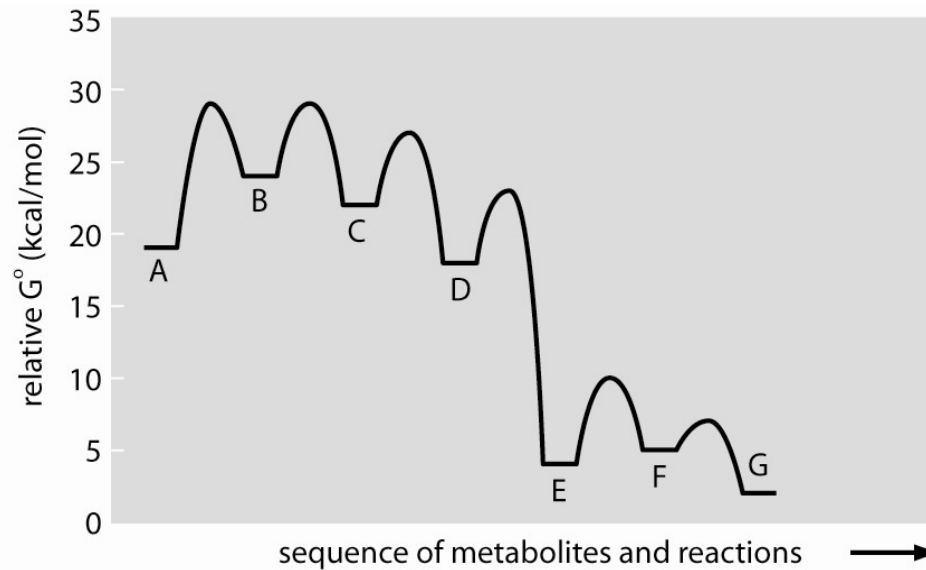
- a) Which reaction(s) is most likely to occur without an enzyme in the cell?
- b) Which reaction(s) is most likely to be catalyzed by an enzyme that simultaneously uses ATP as a substrate?

The energetic trajectory of a series of reactions in a catabolic pathway can be plotted on a graph like that in the figure. The graph shows a hypothetical pathway similar to glycolysis, in which the catabolic intermediates are designated A, B, C, etc. For the following questions, and throughout this exam, explain your answers.



- a) Which reaction(s) is most likely to occur without an enzyme in the cell?
- b) Which reaction(s) is most likely to be catalyzed by an enzyme that simultaneously uses ATP as a substrate?
- c) Which reaction(s) is most likely to be catalyzed by an enzyme that simultaneously uses ADP as a substrate and produces ATP?

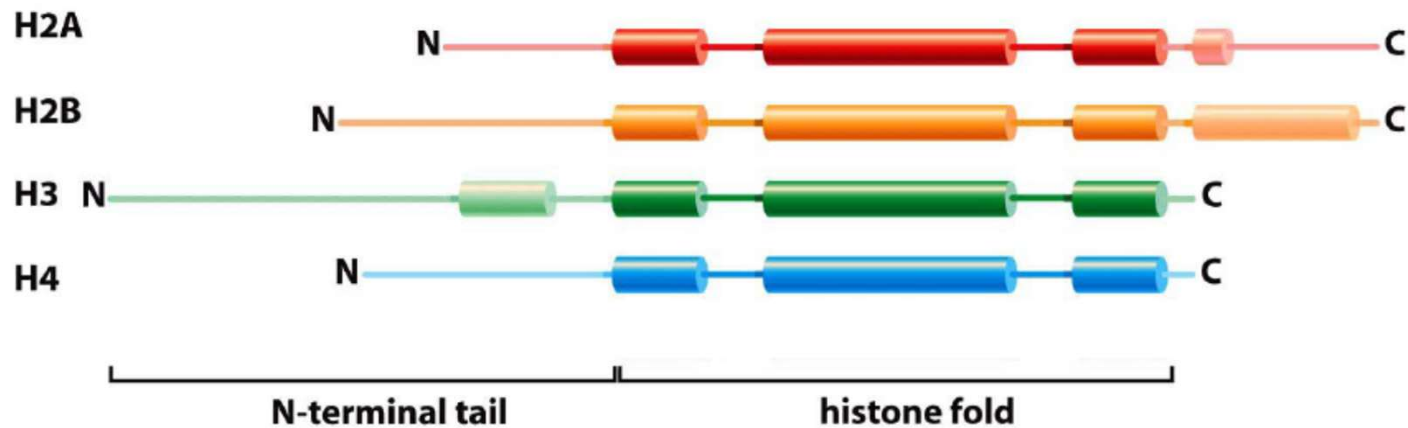
The energetic trajectory of a series of reactions in a catabolic pathway can be plotted on a graph like that in the figure. The graph shows a hypothetical pathway similar to glycolysis, in which the catabolic intermediates are designated A, B, C, etc. For the following questions, and throughout this exam, explain your answers.



- Which reaction(s) is most likely to occur without an enzyme in the cell?
- Which reaction(s) is most likely to be catalyzed by an enzyme that simultaneously uses ATP as a substrate?
- Which reaction(s) is most likely to be catalyzed by an enzyme that simultaneously uses ADP as a substrate and produces ATP?
- Many important molecules in cells are polymers.

Which is thermodynamically more disordered, a polymer or a collection of monomers?

On the basis of your answer, is the ΔG° of a polymerization reaction positive, negative, or zero?



Indicate in the figure the region(s) of the core histone proteins that are the main regions modified by posttranslational covalent modifications which compose the 'histone code'.

Are all four core histones subject to covalent posttranslational modification?

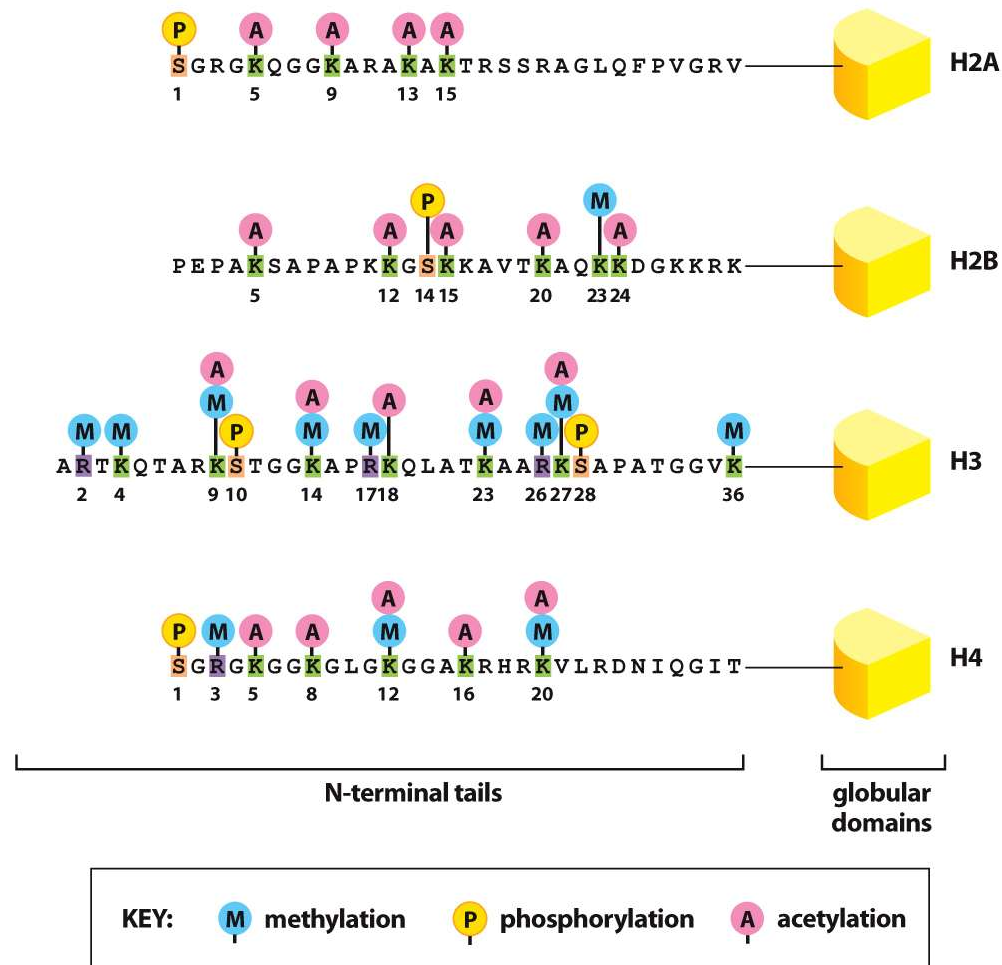
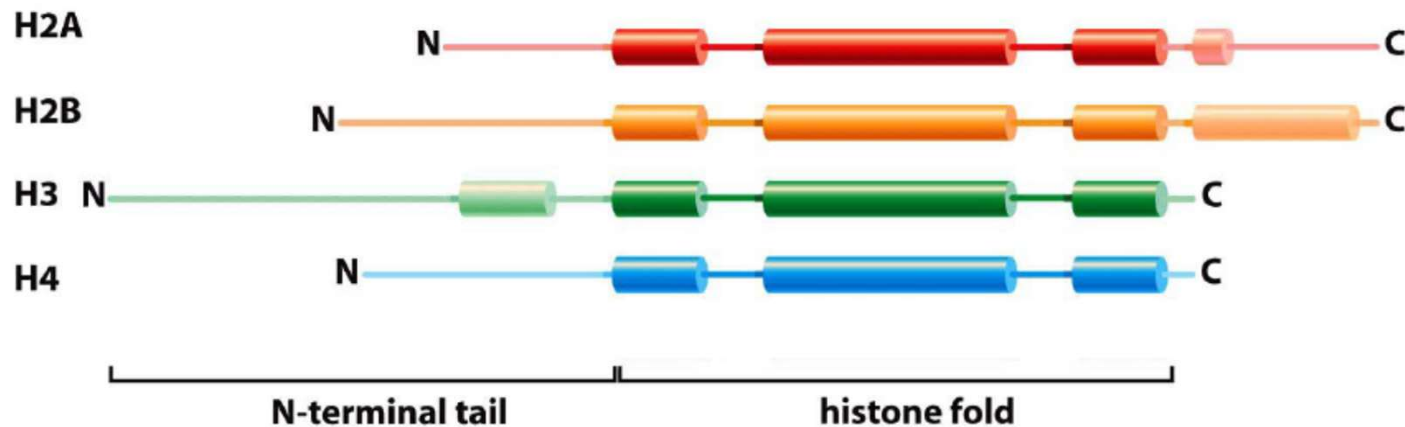


Figure 4-34b Molecular Biology of the Cell 6e (© Garland Science 2015)

Q & A MCB-B



Indicate in the figure the region(s) of the core histone proteins that are the main regions modified by posttranslational covalent modifications which compose the 'histone code'.

Are all four core histones subject to covalent posttranslational modification?

Where are these regions located in the context of a nucleosome?

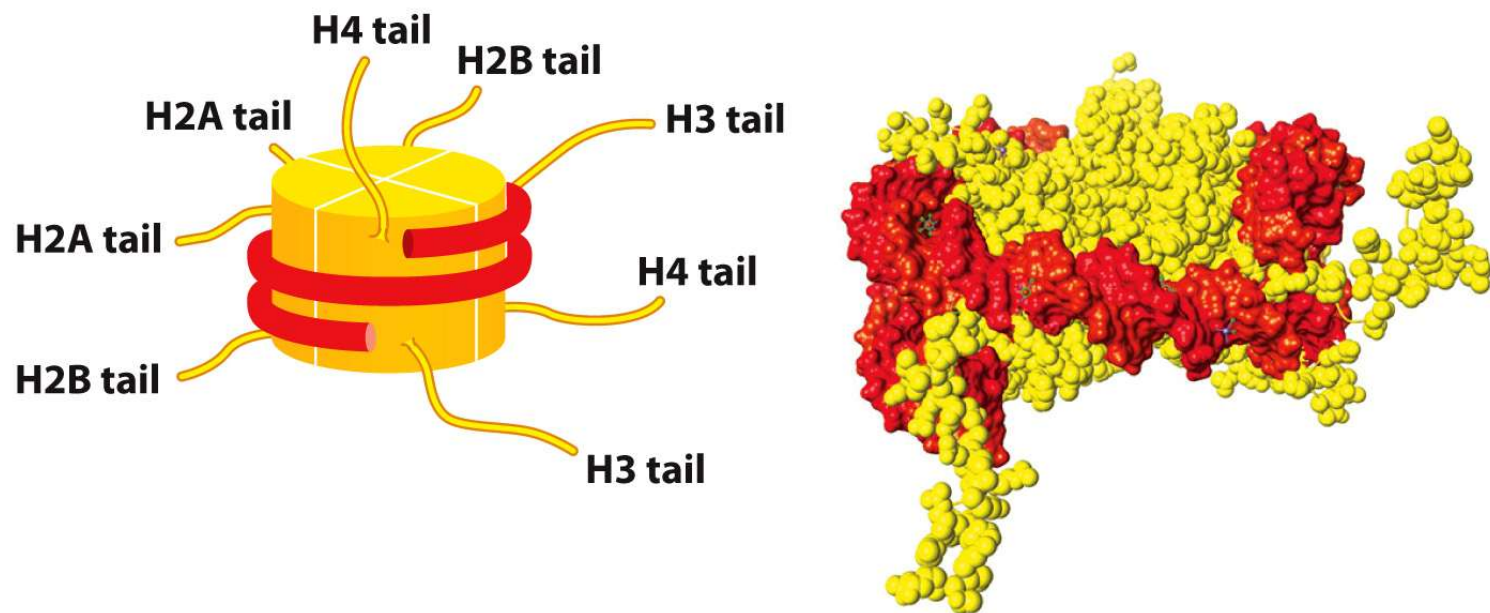


Figure 4-29a Molecular Biology of the Cell 6e (© Garland Science 2015)

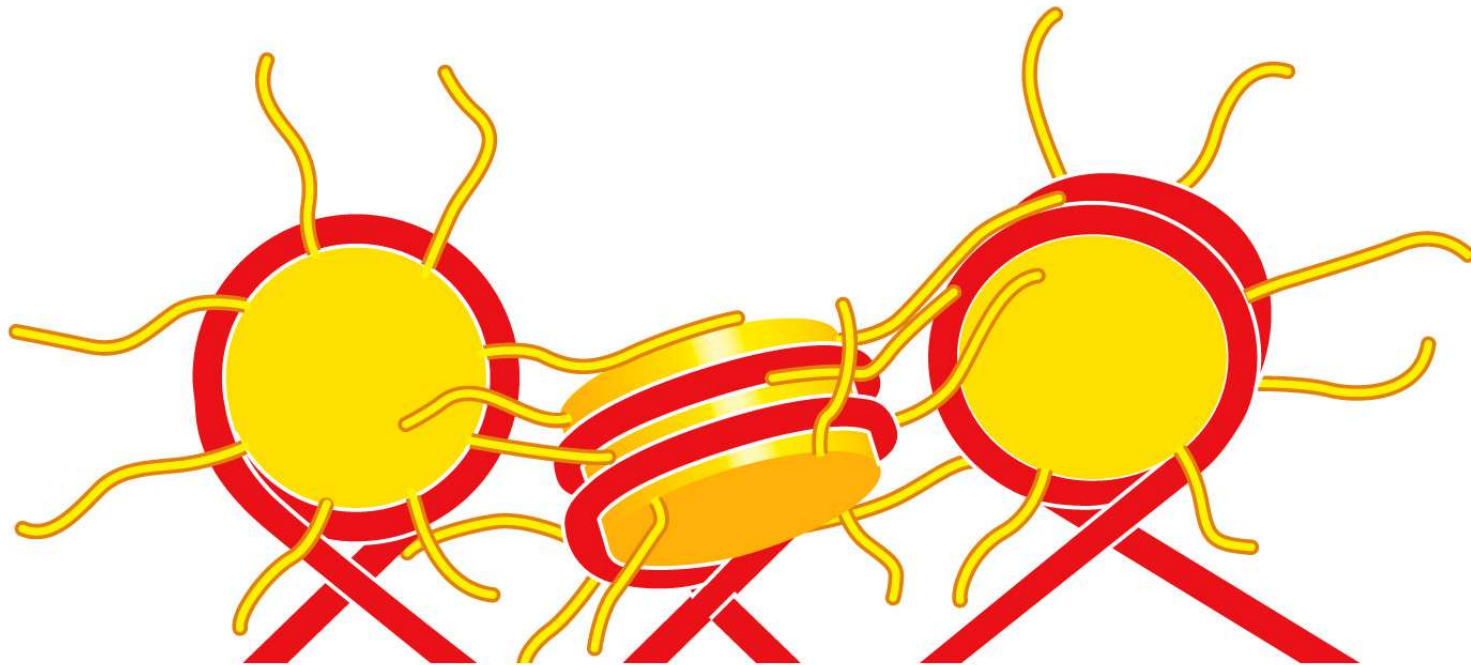
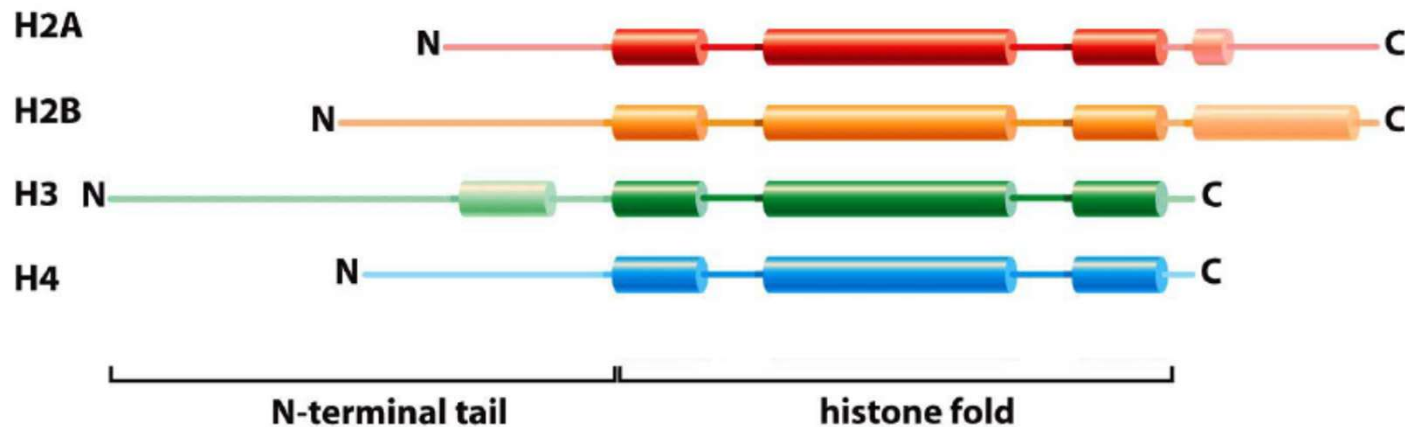


Figure 4-29b Molecular Biology of the Cell 6e (© Garland Science 2015)

Q & A MCB-B



Indicate in the figure the region(s) of the core histone proteins that are the main regions modified by posttranslational covalent modifications which compose the 'histone code'.

Are all four core histones subject to covalent posttranslational modification?

Where are these regions located in the context of a nucleosome?

Can one and the same core histone protein be methylated and acetylated at the same time?

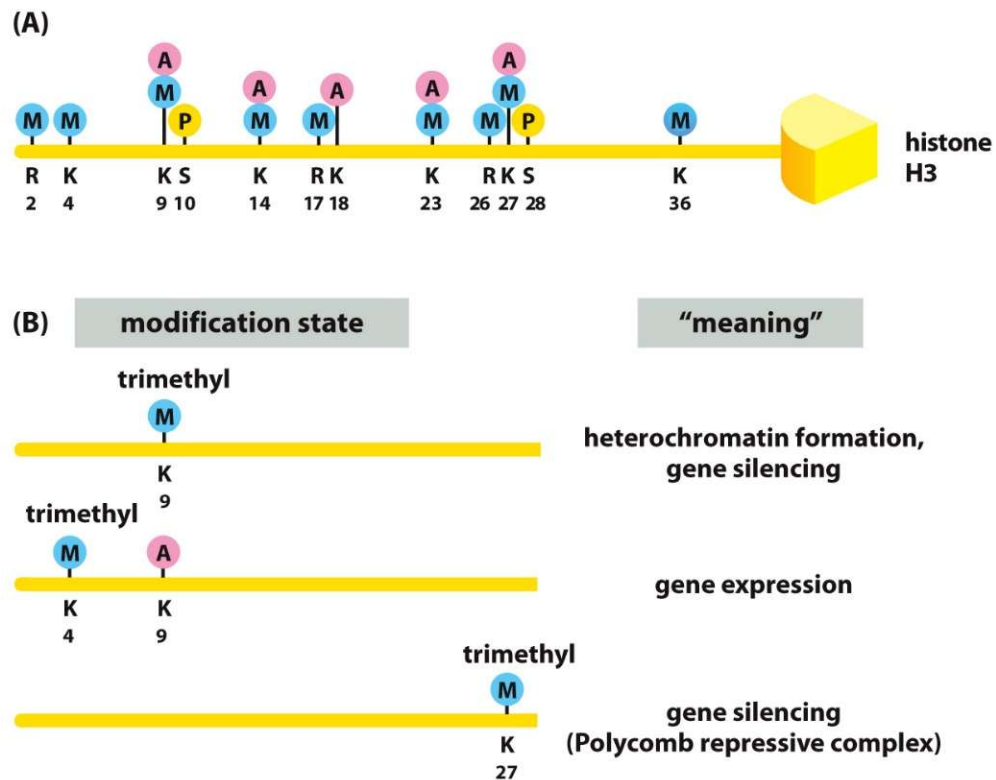


Figure 4-39 Molecular Biology of the Cell 6e (© Garland Science 2015)

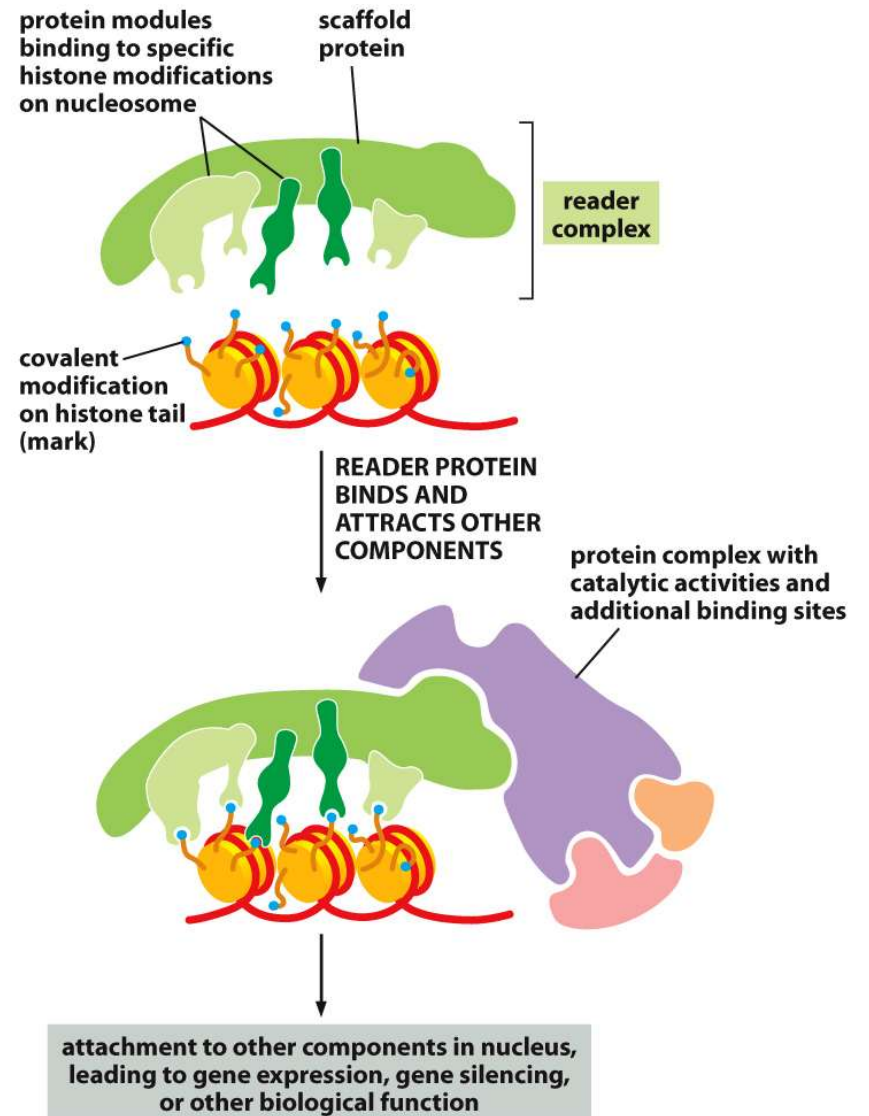


Figure 4-38 Molecular Biology of the Cell 6e (© Garland Science 2015)

In contrast to histone acetylation, which always correlates with gene activation, histone methylation can lead to either transcriptional activation or repression.

How do you suppose that the same type of modification (histone methylation) can mediate different biological outcomes?

In contrast to histone acetylation, which always correlates with gene activation, histone methylation can lead to either transcriptional activation or repression.

How do you suppose that the same type of modification (histone methylation) can mediate different biological outcomes?

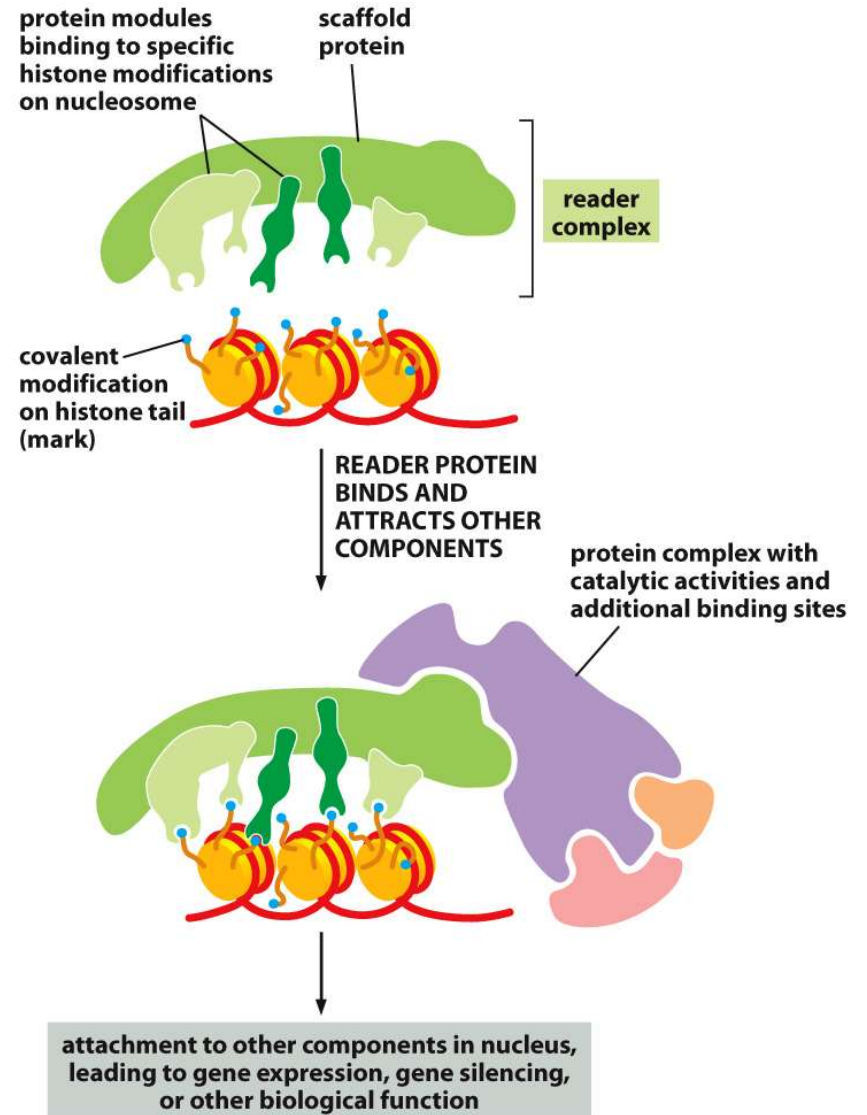


Figure 4-38 Molecular Biology of the Cell 6e (© Garland Science 2015)

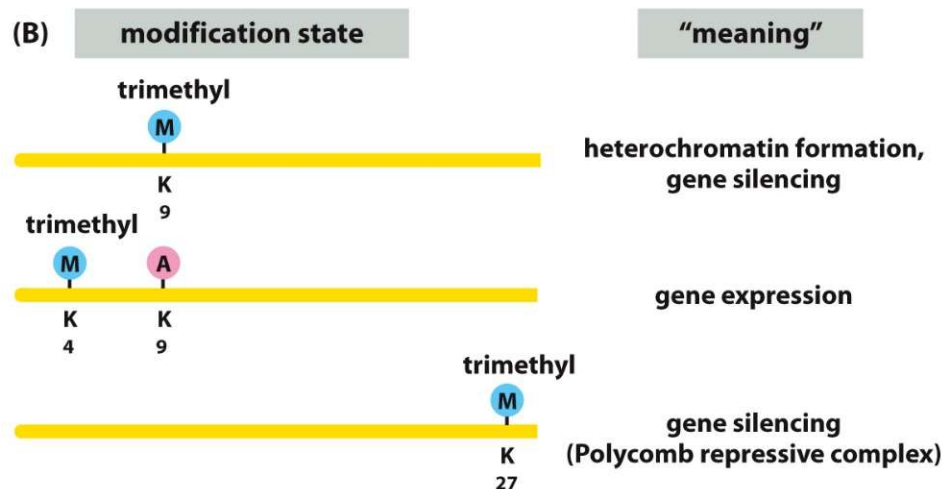


Figure 4-39 Molecular Biology of the Cell 6e (© Garland Science 2015)

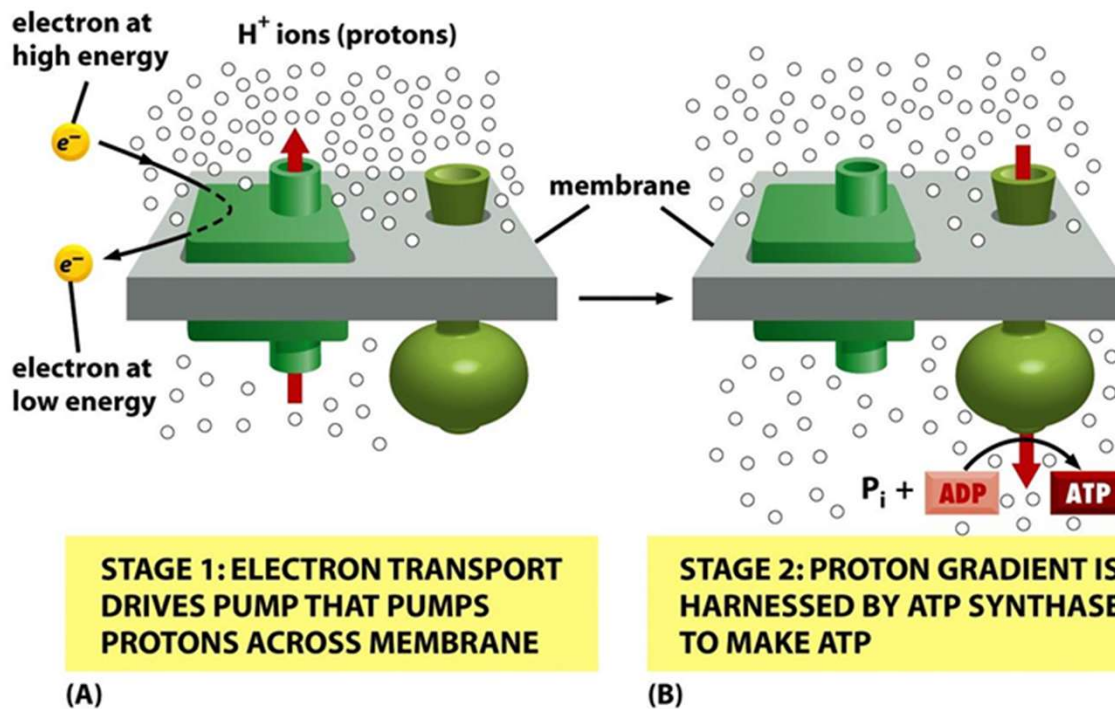
What is chemiosmosis?

How is it related to the production of energy?

What is chemiosmosis?

How is it related to the production of energy?

Chemiosmosis couples electron transport to ATP synthesis via an H^+ gradient



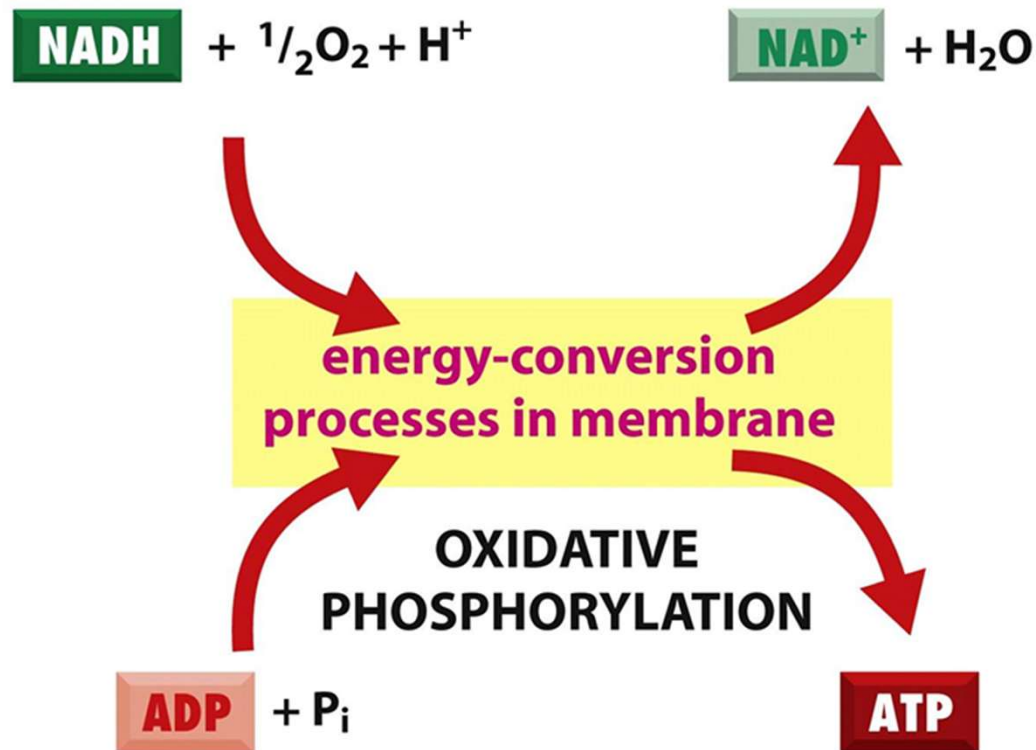
KEY POINTS

- **Stage 1** — electrons pass from high to low energy through membrane proteins that use the released energy to pump H^+ across the membrane.
- **Stage 2** — H^+ flow back down their gradient through ATP synthase, which converts that energy into ATP.
- **The name** — *chemi* = the redox chemistry; *osmosis* = the ion transport across the membrane.

Explain the term oxidative phosphorylation.

Explain the term oxidative phosphorylation.

Oxidative phosphorylation couples NADH oxidation to ATP synthesis



KEY POINTS

- **Oxidation** — $\text{NADH} + \frac{1}{2}\text{O}_2 + \text{H}^+ \rightarrow \text{NAD}^+ + \text{H}_2\text{O}$ releases a large amount of free energy.
- **Phosphorylation** — $\text{ADP} + \text{P}_i \rightarrow \text{ATP}$ requires energy.
- **Coupling** — energy-conversion processes in the inner membrane (the H^+ gradient) link the two.